

Computer aided mathematics and visualization
Practice (2025)

1. Let us consider the following surface:

$$\begin{aligned}x(u, v) &= u - \frac{u^3}{3} + uv^2 \\y(u, v) &= v - \frac{v^3}{3} + vu^2 \\z(u, v) &= u^2 - v^2 \\u &\in [-25, 25], \quad v \in [-25, 25]\end{aligned}$$

Draw the surface! Draw the point corresponding to $u = 10$ and $v = 15$ along with the corresponding isocurves. Draw the normal vector of the surface at P .

2. Let us consider the surface

$$z = \sqrt{1 - x^2 - 0.5y^2}.$$

Draw the surface! Draw the point on the surface corresponding to $x = 0.5$ and $y = 0.2$.

3. Let us consider the surface

$$\sin(x) + \frac{\cos(y)}{x} - z = 0.$$

Draw the surface! Draw the intersection of the surface with the xy plane.

4. Let us define the following 3 planes:

$$x + y - z = 0, \quad x - 2y + 3z = 4, \quad 2x - 0.5y + 4z = -2.$$

Draw them with different colors!

5. Let us have curves

$$\begin{aligned}p(u) &= (1 - u) P_1 + u P_2 \\r(u) &= (1 - u) R_1 + u R_2 \\u &\in [0, 1]\end{aligned}$$

where $P_1 = (0, 0, 0)$, $P_2 = (0, 1, 1)$, and $R_1 = (1, 0, 1)$, $R_2 = (1, 1, 0)$. Let us define the parametric surface as follows:

$$\begin{aligned}s(u, v) &= (1 - v) p(u) + v r(u) \\u &\in [0, 1], \quad v \in [0, 1]\end{aligned}$$

Draw the given two curves and the given surface on the same figure.

6. Draw a polynomial curve of degree 4 that goes through the points $(10, 20)$, $(20, 40)$, $(40, 40)$, $(50, 20)$, $(20, 10)$ when the parameter is 0, 1, 2, 3, and 4 respectively. Draw the tangent vector of the curve when $t = 0.5$
7. Let us define the points $P_1 = (-2, -2)$, $P_2 = (4, 0)$, $P_3 = (6, -2)$, $P_4 = (10, 2)$. Draw the Hermite arc that goes through these points at the parameter values $-1, 0, 2, 3$ respectively. Draw the tangent vector of the curve when $t = 2$.

8. Let us define the points $P_1 = (-2, -2)$, $P_2 = (6, -2)$, $P_3 = (10, 2)$, and the vector $\mathbf{v} = (6, -4)$. Draw the Hermite arc that goes through these points at the parameter values 0, 1, 1.5, and whose tangent vector at 0 is vector $\mathbf{v}$.
9. Let us define the points $P_1 = (-2, -2)$, $P_2 = (6, -2)$, and vectors $\mathbf{v}_1 = (6, -4)$ and $\mathbf{v}_2 = (4, 4)$. Draw the Hermite arc that goes through these points at the parameter values 0, 1, and whose tangent vector at 0 is $\mathbf{v}_1$ and at 1 is $\mathbf{v}_2$.
10. Draw a Bézier curve with the control points $(10, 20)$, $(20, 40)$, $(40, 40)$, $(50, 20)$, $(20, 10)$. Draw its tangent vectors at its beginning and at its end.